TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

MAY EXAMINATION

BMIT3084 ENTERPRISE NETWORKING

THURSDAY, 7 MAY 2026 TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INFORMATION SECURITY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INTERNET TECHNOLOGY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

This question paper consists of 4 questions on 7 printed pages.

Question 1

You are appointed as a network administrator to analyse, demonstrate, and configure different types of static routes for a network topology where Internet Protocol version 4 (IPv4) addressing is implemented on all devices, as illustrated in **Figure 1-1**.
Assume that a **fully specified default static route** has been preconfigured on **ROUTER_C**, and a **default static route** has also been preconfigured on **ROUTER_A** to forward traffic to the **ISP**. Additionally, **standard static routes** have been preconfigured on the **ISP** to route packets to **PC1** and **PC2**.

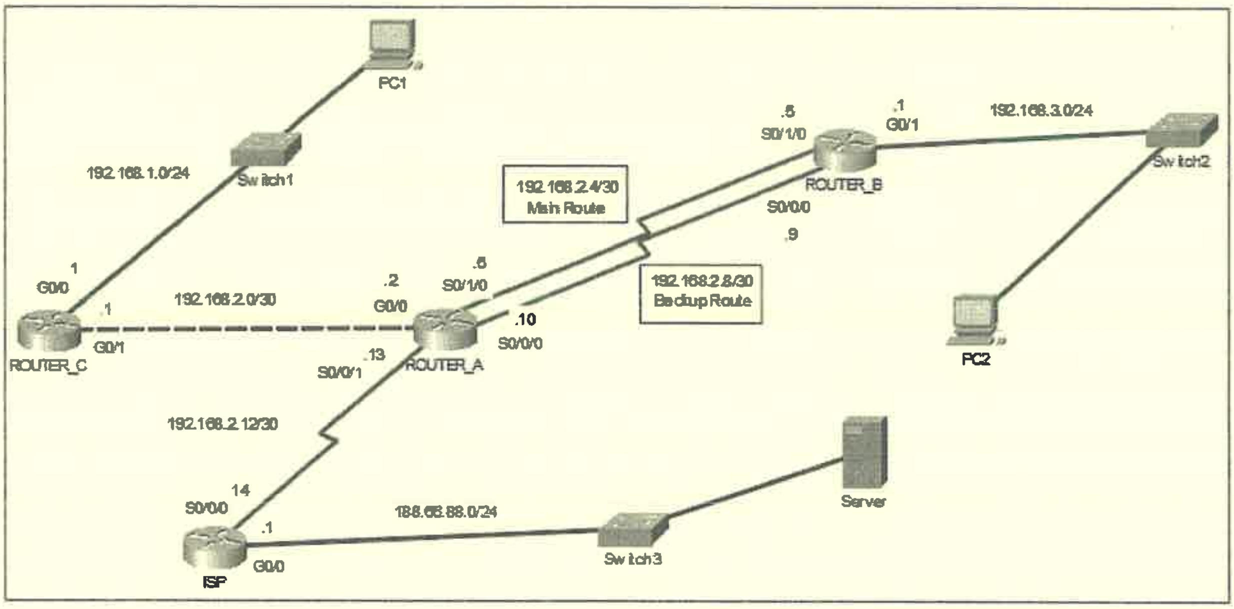

Figure 1-1: A network topology

a) With reference to **Figure 1-1**, answer the following questions.

(i) **ROUTER_B** requires a primary default route and a backup (floating) default route to forward traffic to **ROUTER_A**. Explain how the concept of **Administrative Distance (AD)** is manipulated to achieve this redundancy, and describe the router's exact behaviour when the primary link fails. (8 marks)

(ii) Compare the operational differences between a **standard static route** (using only a next-hop IP address) and a **fully specified static route** (using both an exit interface and a next-hop IP address) when **ROUTER_A** forwards packets to the **PC1** network. (4 marks)

(iii) Explain **TWO (2)** reasons why a **fully specified standard static route** need to be configured in **ROUTER_A**. (4 marks)

b) With reference to **Figure 1-1**, assume that **Open Shortest Path First version 2 (OSPFv2)** with Process ID **88** and Area **0** is being implemented to establish dynamic routing between **ROUTER_A**, **ROUTER_B**, and **ROUTER_C**. A **static default route** is pre-configured on **ROUTER_A** to forward traffic to the **ISP**. Apply **OSPFv2** theoretical concepts to answer the following:

Question 1 b) (Continued)

(i) As shown in **Figure 1-1**, the connection between **ROUTER_A** and **ROUTER_C** uses the subnet mask **255.255.255.252**. Apply the mathematical rule for deriving the **OSPFv2** wildcard mask from the given subnet mask. Demonstrate your calculation and explain how **OSPFv2** applies this resulting wildcard mask to activate the correct interface for **Area 0**. (5 marks)

(ii) **ROUTER_A** uses the **default-information originate command** to propagate its static default route into the **OSPFv2 Area 0** domain. Explain the **TWO (2)** technical requirements that must be met for **ROUTER_A** to successfully advertise this route. (4 marks)

[Total: 25 marks]

Question 2

a) **Access Control Lists (ACLs)** are used to filter network traffic. Based on **Figure 2-1**, answer the following questions. Assume all hosts and routers can communicate with each other.

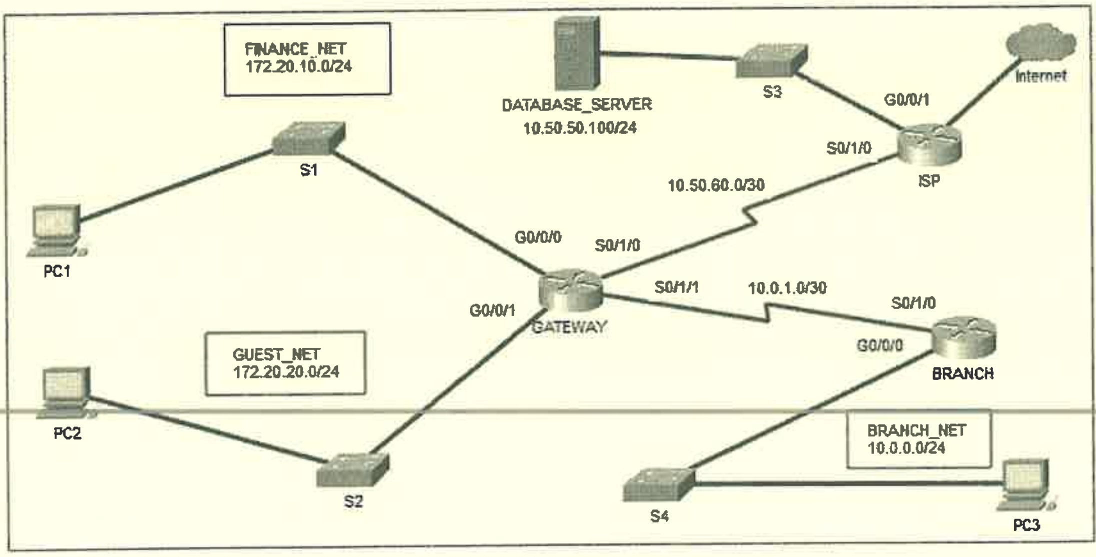

Figure 2-1: A network topology

(i) Write an **Extended Named ACL** called **SECURE_ACCESS** to allow only **even-numbered** host IP addresses from **FINANCE_NET** network to access the **DATABASE_SERVER** via **SQL** (Port 1433). Allow **GUEST_NET** network to access the Internet via **HTTPS** (443) but deny them access any internal **BRANCH_NET** network addresses. Explicitly deny all other traffic. Use port number for services and suitable keyword(s) in your ACL. Specify the router, interface(s) and direction for the most efficient placement. (17 marks)

(ii) Explain the concept of "**Implicit Deny**" in ACLs and how it affects traffic if a "**permit any**" statement is omitted. (4 marks)

Question 2 (Continued)

b) *A malicious actor succeeds in connecting a rogue device to the internal LAN switch of your company. This device begins responding to DHCP Discover messages sent by legitimate employees' laptops.*

Based on this scenario, identify this specific network attack and analyse **ONE (1)** way the attacker can configure the rogue server's responses to manipulate or disrupt client traffic.

(4 marks)

[Total: 25 marks]

Question 3

OSPFv2 and static routing are implemented in the respective routers as shown in **Figure 3-1** network topology. All devices can communicate with each other. With reference to **Figure 3-1**, answer the following questions.

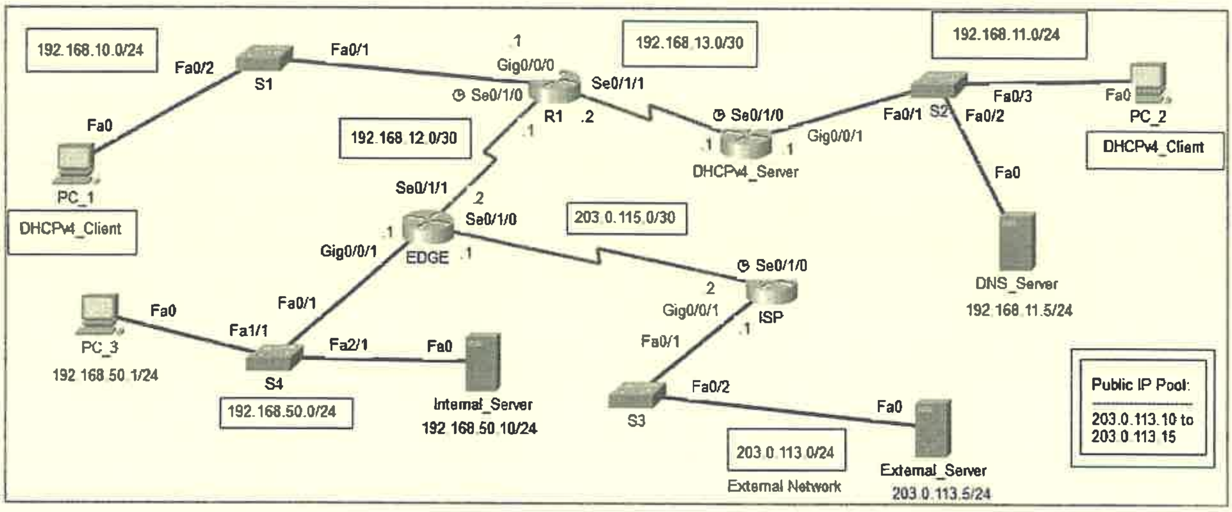

Figure 3-1: A network topology

a) Examine the requirement for the **Internal_Server**, to be directly reachable from the external network using the public IP address **203.0.113.254/24**. State the type of network service required to fulfil this requirement, and describe the step-by-step process when an external user sends an HTTP request to this server. In your explanation, clearly identify the **Inside Local** and **Inside Global** IP addresses associated with this server.

(5 marks)

b) The network shown in **Figure 3-1** requires the **EDGE** router to facilitate Internet access for the **192.168.50.0/24** network. You are tasked with applying the theoretical framework of **Port Address Translation (PAT)** using a public pool named **NAT_POOL** (range: **203.0.113.10** to **203.0.113.15**).

Complete **Table 3-1** by identifying the theoretical values based on the given topology and specific requirements outlined in the table.

(Hint: Do not provide Cisco IOS commands. Provide only the theoretical values, terminology, or logic requested.)

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Question 3 b) (Continued)

Table 3-1: PAT Operational Documentation

Component Category	Specific Requirement	Theoretical Value
Interface Selection	Inside Interface	
	Outside Interface	
Address Classification	Inside Local Address Range	
	Inside Global Address Pool	
Traffic Filtering	ACL Wildcard Mask	
	Purpose of the ACL	
Translation Method	Internet access for all hosts on the 192.168.50.0/24 network	

(8 marks)

c) **Dynamic Host Configuration Protocol version 4 (DHCPv4)** configurations are to be configured in **DHCPv4_Server** router to automatically lease IP addresses to **PC_1** (Left_Network: 192.168.10.0/24) and **PC_2** (Right_Network: 192.168.11.0/24). The requirement states that the first 5 IP addresses of both networks must be excluded, and a relay agent must be configured on router **R1** to forward requests to the DHCP server at 192.168.13.1. However, after applying the configuration shown in **Table 3-2**, the hosts are either failing to receive IP addresses or are receiving incorrect parameters.
Apply your knowledge of **DHCPv4** command syntax and operation to troubleshoot the script. Identify **FOUR (4)** configuration errors in the provided script. For each error, state the operational impact (why it fails) and provide the exact corrected command. Document your answers using **Table 3-3**.

Table 3-2: Configuration Commands

Router name	Configurations
DHCPv4_Server	# ip dhcp excluded-address 192.168.11.5 # ip dhcp pool Right_Network # network 192.168.11.0 255.255.255.0 # default-router 192.168.11.1 # dns-server 192.168.11.5 # ip dhcp excluded-address 192.168.10.1 192.168.10.5 # ip dhcp pool Left_Network # network 192.168.10.0 0.0.0.255 # default-router 192.168.10.0 # dns-server 192.168.11.5
R1	# interface g0/0/0 # ip dhcp helper 192.168.13.1

Question 3 c) (Continued)

Table 3-3: Correction Table

Error Location / Incorrect Command	Operational Impact (Why it fails)	Corrected Command

(12 marks)

[Total: 25 marks]

Question 4

a) GlobalConnect Logistics, a medium-sized shipping company with a Headquarters (HQ) and four regional branch offices, is implementing a new real-time tracking system that demands high availability. While the company has a moderate budget that allows for additional hardware at the central hub, they cannot justify the high cost of interconnecting all branch offices, especially since 95% of data traffic is sent directly to the HQ and branches rarely communicate with one another. Furthermore, the infrastructure must ensure that the HQ remains connected even if a primary service provider fails. Analyse these requirements to determine which **network topology** best balances the need for core redundancy with existing traffic patterns and budget constraints. In your answer, explain why this topology is the best choice and how it addresses each of the company's specific constraints.
(Hint: You must select a Dual-homed Hub-and-Spoke, Hub-and-Spoke, Full Mesh, Partial Mesh, or Point-to-Point topology) (9 marks)

b) GlobalConnect Logistics is upgrading its network to support a new suite of real-time applications, including Voice over Internet Protocol (VoIP) for dispatchers, high-definition video conferencing for management, and bulk data transfers for inventory synchronisation. The company currently faces significant congestion on its Wide Area Network (WAN) links, causing dropped voice calls and sluggish database updates during peak hours. Given that the company requires a strict guarantee of low-level latency for voice traffic while ensuring that mission-critical database applications receive a fair, minimum bandwidth allocation without being 'starved' by large file transfers, analyse the standard **Quality of Service (QoS) algorithms** to determine which single algorithm is best suited for this environment. In your response, explain why this algorithm is the superior choice for a multi-traffic environment and how it specifically addresses the conflict between real-time voice needs and bulk data throughput.
(Hint: You must select a First Out (FIFO), Low Latency Queuing (LLQ), Class-Based Weighted Fair Queuing (CBWFQ), or Weighted Fair Queuing (WFQ) algorithm) (9 marks)

c) GlobalConnect Logistics initially deployed a **GRE (Generic Routing Encapsulation) tunnel** to connect its regional offices across the public Internet. However, a recent security audit flagged this infrastructure as high risk for the company's real-time tracking data and financial records. Analyse the technical characteristics of a **GRE-only tunnel** and determine why it is an insufficient choice for transporting sensitive data over an untrusted network.

Question 4 c) (Continued)

In your answer, identify which specific security framework should be integrated with the **GRE-only tunnel** to rectify these vulnerabilities and explain how this framework provides confidentiality and origin authentication. (7 marks)

[Total: 25 marks]